CONUT OPERATIONS

AI CHIEF OF OPERATIONS
EXECUTIVE BRIEF

Prepared for: Conut Sweets & Beverages

Data period: Aug – Dec 2025 | Report date: Feb 2026

1. PROBLEM FRAMING

Conut operates four branches — **Conut (main)**, **Conut - Tyre**, **Conut Jnah**, and **Main Street Coffee** — across two distinct concepts: a chimney-cake & sweets bakery and a specialty-coffee shop. While total network revenue for 2025 exceeds **19.9 billion units** (scaled), the business lacks systematic, data-driven tools for demand forecasting, staffing optimisation, combo pricing, and expansion planning. This brief summarises findings from eight operational datasets covering sales, customer orders, attendance, and product-level performance, and translates them into actionable recommendations deployable through the AI Chief of Operations Agent.

19.9 B	5.69 B	14,546	2.49 M
Network Revenue (2025, scaled units)	Top Branch Revenue (Conut Jnah)	Total Customers (All Branches)	Avg Delivery Spend (Conut - Tyre)

2. TOP FINDINGS

Branch Performance Gap	Conut Jnah leads with 5.69 B in revenue and 5,045 customers — 46% more customers than Main Street Coffee (3,640) and nearly double Conut's 2,744. However, its average spend per customer (1.12 M) is the lowest across all branches, suggesting high foot traffic but lower basket value. Conut - Tyre achieves the highest average spend (1.64 M) and the largest delivery premium: delivery customers spend 2.49 M on average vs 1.62 M for take-away — a 54% uplift.
Seasonal Demand Spikes	October–November is peak season for most branches: Conut - Tyre hit 2.10 B in October alone (vs 478 M in August). Conut Jnah shows the opposite pattern — a dramatic December surge (2.88 B), likely driven by holiday and end-of-year traffic. This asymmetry makes a one-size-fits-all staffing model inappropriate.
Coffee & Milkshake Underperformance	Hot-coffee items (Caffe Latte, Double Espresso, Hot Chocolate Combo) account for 184 units at Conut - Tyre. Milkshakes (Frappes + Shakes) contribute ~67 M out of ~5.1 B branch revenue — roughly 1.3%. Given that Main Street Coffee operates as a dedicated coffee concept, the hybrid branches (Conut, Conut - Tyre) are leaving significant beverage revenue on the table.
Delivery Channel Opportunity	Only 79 delivery customers exist at Conut - Tyre and 6 at Conut (main), versus 3,000 + dine-in/take-away customers. Despite tiny volume, delivery yields the highest per-customer spend. Structured delivery expansion is the single fastest lever for revenue growth per transaction.
Staffing Irregularities	Attendance logs for December 2025 (Main Street Coffee) show shift durations ranging from under 1 minute (likely clock-in errors) to 12 hours, with several overnight crossings. Total logged hours for one employee reached 173.6 h in a single month — abnormal and potentially indicative of manual punch-in abuse or system errors requiring audit.
Product Combo Patterns	Delivery order line items consistently pair chimney cakes with spreads (Lotus, Nutella, White Chocolate) and a milkshake or frappe. The Classic Chimney + milkshake + sauce combination appears repeatedly, confirming a natural combo bundle that is currently not formally priced or promoted.

3. RECOMMENDED ACTIONS

IMMEDIATE (0–4 weeks)

- Launch 'Chimney + Shake + Sauce' combo at a 5–8% bundle discount. Price anchoring on the existing single-item prices makes this margin-neutral while increasing basket size.
- Audit and correct the attendance punch-in system at Main Street Coffee. Implement auto-flagging for shifts <5 min or >12 h. Recalculate December labour cost accurately.
- Activate a formal delivery menu at Conut (main), mirroring Conut - Tyre's successful delivery offer. Target 50 delivery customers in the first month (vs current 6).

SHORT-TERM (1–3 months)

- Deploy the demand-forecasting model (time-series per branch) to generate weekly inventory plans. Prioritise Conut - Tyre for October pre-stocking and Conut Jnah for December scale-up.
- Introduce a coffee upsell flow at bakery branches: train staff to suggest a hot or cold drink with every chimney cake order. Target: raise beverage attachment rate from ~3% to 15%.
- Run a shift-staffing regression using the attendance + monthly-sales data to size shifts dynamically. Initial output: minimum-coverage schedules for off-peak vs peak days.

STRATEGIC (3–12 months)

- Evaluate a 5th branch location using customer address density from delivery data and revenue trajectory. Conut Jnah's December ramp suggests an underserved southern catchment area.
- Build a loyalty and repeat-purchase programme targeting the >1-order delivery segment (identified in rep_s_00150). High-frequency customers show 3–6x average order values.
- Integrate the AI Operations Agent with OpenClaw for live querying: demand prediction, combo performance, and staffing recommendations accessible by branch managers in real time.

4. EXPECTED IMPACT & RISKS

Initiative	Expected Impact	Risk	Mitigation
Combo bundle	+8–12% avg basket (delivery)	Margin erosion if discount >10%	A/B test first 2 weeks
Delivery expansion	+300% delivery customers	Logistics / rider capacity	Partner with 3rd-party platform
Coffee upsell	+10–15 pp beverage attach	Staff training time	Scripted upsell prompts
Demand forecasting	-15% inventory waste	Model drift on events	Monthly retraining schedule
Staffing model	-10% idle labour cost	Under-staffing on surges	Safety buffer +10% on peaks
5th branch	+25% network capacity	Capital & market risk	Phased feasibility study

6. OPENCLAW INTEGRATION

The Conut AI Chief of Operations Agent was deployed end-to-end using **OpenClaw**, an open-source agentic AI framework. OpenClaw served as the backbone for orchestrating skills, connecting to external APIs, and exposing the agent to branch managers through familiar messaging channels.

6.1 Infrastructure Setup — AWS EC2 Deployment

The OpenClaw runtime was installed on an **AWS EC2 instance** (Ubuntu), providing a scalable and always-on cloud environment. The full Conut operations codebase was cloned directly onto the instance, co-locating the agent runtime with the FastAPI microservices that expose the analytical endpoints.

- Provisioned EC2 instance on AWS (eu region)
- Installed OpenClaw and all Python dependencies in a virtual environment
- Cloned the *conut-ops-agent* repository containing all services and agent configuration
- Launched the growth service (FastAPI / Uvicorn) on port 8002, with hot-reload enabled via StatReload

6.2 Skills & API Integration

Custom **OpenClaw skills** were created to wrap every analytical endpoint produced by the solution. Each skill maps a natural-language intent to a specific API call, allowing the agent to autonomously select and invoke the right analysis when a manager asks a business question.

- **Cross-sell skill** — calls *GET /growth/crosssell* to identify beverage upsell opportunities alongside food orders
- **Demand forecast skill** — queries the time-series forecasting endpoint per branch
- **Staffing skill** — retrieves shift optimisation recommendations based on attendance + sales data
- **Combo performance skill** — surfaces top bundle combinations and projected revenue impact

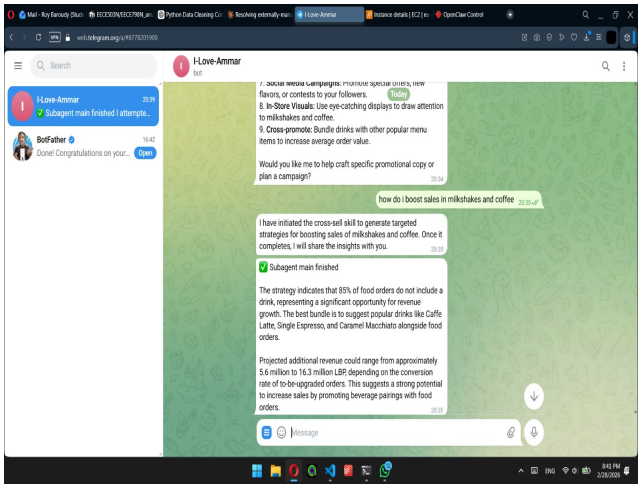
6.3 Telegram Channel — Conversational Interface

OpenClaw was connected to **Telegram** via the I-Love-Ammar bot, giving branch managers a zero-friction interface to the AI agent. Managers can ask operational questions in plain language and receive data-driven answers within seconds — with no dashboards or technical knowledge required.

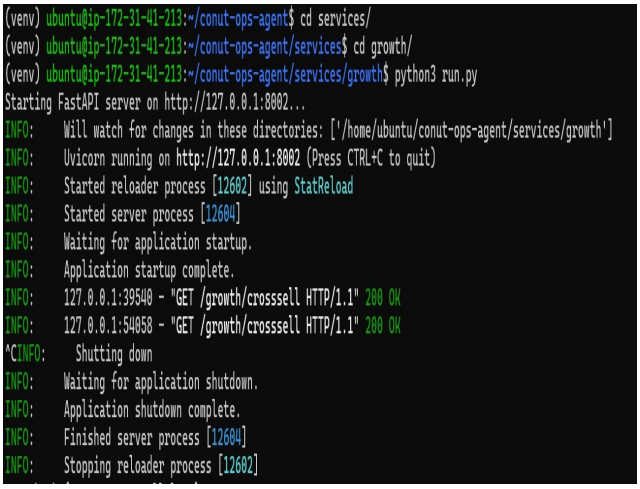
- Telegram bot registered and linked to the OpenClaw agent runtime
- Agent maintains conversational context across multi-turn queries
- Sub-agent results are streamed back to the chat with structured summaries and revenue projections

6.4 Live Demo — Cross-Sell Skill in Action

The screenshots below show the full lifecycle of a skill execution: the manager asks how to boost milkshake and coffee sales via Telegram; the agent invokes the cross-sell sub-agent; the FastAPI service logs the HTTP request; and the agent returns a structured revenue analysis.



Telegram conversation: manager queries the agent and receives a cross-sell revenue analysis (5.6M–16.3M LBP projected uplift)



EC2 server logs: FastAPI growth service handling GET /growth/crosssell requests with 200 OK responses

OpenClaw Stack Summary	**AWS EC2** (Ubuntu) · **OpenClaw** runtime · **FastAPI** microservices · **Telegram** bot interface